

DRY-RUN GUIDE | 04_gcd.cpp
Source: cpp\2nd_belt\04_gcd.cpp

CORE BEHAVIOR

Takes absolute values and applies Euclid's remainder algorithm until the second value is zero.

REPRESENTATIVE INPUT / CALL

Input: '48 18'.

STEP TRACE

(48,18)->(18,12)->(12,6)->(6,0); the remaining first value is the GCD.

EXPECTED OUTPUT / RETURN

'6'

ACTUAL SOURCE EXCERPT

Verbatim lines from the source; long lines wrap. Source files were not modified.

```
001 | /*
002 | Question:
003 | Write a program to calculate and print the Greatest Common Divisor (GCD)
004 | of two numbers. Note that if either number is zero, print the other number.
005 |
006 | Input Format:
007 | The input consists of two integers A and B.
008 |
009 | Output Format:
010 | Print the GCD of the two numbers.
011 | */
012 |
013 | #include <iostream>
014 | #include <cstdlib>
015 | using namespace std;
016 |
017 | int main() {
018 |     int a, b;
019 |     cin >> a >> b;
020 |
021 |     a = abs(a);
022 |     b = abs(b);
023 |
024 |     while (b != 0) {
025 |         int remainder = a % b;
026 |         a = b;
027 |         b = remainder;
028 |     }
029 |
030 |     cout << a;
031 |
032 |     return 0;
033 | }
034 |
```